Element E: Application of STEM Principles and Practices

Student E, a student in our engineering class, needs a way to get rid of wasps, hornets, and other insects that enter his room at night when he sleeps. The student's bedroom has a serious insect problem. The insects make the space noisy and affect the students’ ability to rest.

Many insects are drawn to light, especially smaller insects like mosquitoes and some wasp species. By using the right kind of light in a small trap, we can attract more insects. New technology, like small UV, ultra-violet, lights can attract insects to the trap (*Efficiency of UV LEDs at attracting insects, 2025)*.

To ensure that our prototype meets our stakeholders’ needs, we ensured that the prototype was smaller than 14 cubic inches by measuring its size with a ruler and its diameter with a calliper. We measured that the prototype did not exceed 1 pound in weight to ensure that this design is easily usable and portable. To ensure that the prototype was easily assembled within 10 minutes, we used a stopwatch to measure the time it took to assemble the prototype.

Sticky traps can be harmful to humans, pets, and the environment due to the presence of strong adhesives, added insecticides, and chemical attractants. The glue used can cause skin irritation, allergic reactions, or injury if it comes into contact with skin, fur, or feathers. Some traps also contain toxic substances like pyrethroids or boric acid, which can be poisonous if ingested by pets or children, leading to symptoms such as vomiting, tremors, or even seizures. Environmentally, these traps can unintentionally catch beneficial insects, small animals, or birds, causing suffering or death and disrupting local ecosystems. Improper disposal can also lead to pollution and long-lasting environmental damage (*“Glue Traps.”, 2025)*.

Following the initial testing, we consulted with experts, Dr. M and Dr.H, to gather feedback and improve the design. Dr. M is a beekeeper with extensive experience dealing with insects. His expertise in managing insect problems in his beekeeping makes him uniquely qualified to assess our prototype's effectiveness. Dr. H is very knowledgeable about insects, a subject that was vital in her previous positions. She has also taught biology for a considerable time and has a strong interest in animals, particularly insects. Their combined knowledge and practical experience provided valuable insights into insect behavior and effective control methods.

During the meeting, Dr. H noted that the specific species of insects we are targeting are not attracted to light. This observation challenged our initial design concept, which relied on light attraction. Both experts suggested modifications to better contain and attract the insects and prevent them from escaping or being driven to other areas. They emphasized the importance of ensuring that the insects, once inside the trap, could not easily escape or be diverted to unintended locations, aligning with our design requirement of containing the insects. They also suggested that it might be better to address the source from which the insects are entering instead of trying to design a solution for dealing with the insects inside.

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